Chapter Contents

Chapter 19 - Further Numerical Methods

Chapter 18 introduced explicit Euler's method. Euler uses one slope at the left endpoint of each step, so it is simple but often requires a small step size for useful accuracy.

This chapter develops methods that use more slope information:

- modified Euler samples both ends of a proposed step
- classical Runge–Kutta samples four carefully chosen stage points
- Adams–Bashforth–Moulton uses several previous slopes
- Milne's method combines a history-based predictor with a Simpson-type corrector

We continue to study:

$$y' = f(x, y), \qquad y(x_0) = y_0,$$

on an equally spaced grid:

$$x_n = x_0 + nh.$$

The main questions are:

- what information does each method sample?
- why does a predictor need a corrected endpoint slope?
- what do the four RK4 stages represent?
- what does the order of a method tell us about global error?
- why do multistep methods need start-up values?
- how are old slopes indexed in ABM4 and Milne's method?
- what can a large predictor–corrector gap warn us about?
- how should accuracy, stability, storage, and computational cost be balanced?

From One Slope To Several Slopes

The Goal Of This Block

The goal is to establish notation and see how one-step and multistep methods improve on Euler in different ways.

Recall Euler's Limitation

Euler's update is:

$$y_{n+1} = y_n + hf(x_n, y_n).$$

Define the current slope:

$$f_n = f(x_n, y_n).$$

Then Euler becomes:

$$y_{n+1} = y_n + hf_n.$$

The method assumes that the left-endpoint slope f_n represents the whole interval $[x_n, x_{n+1}]$. If the slope changes noticeably across that interval, the approximation bends too late.

Two Ways To Gather More Information

A **one-step method** calculates y_{n+1} using information generated from the current point (x_n, y_n). Modified Euler and RK4 are one-step methods.

A **multistep method** also uses saved information from earlier points such as:

$$f_{n-1}, \qquad f_{n-2}, \qquad f_{n-3}.$$

ABM4 and Milne's method are multistep methods.

The practical difference:

one-step methods spend more work inside the current interval; multistep methods reuse work stored from previous intervals.

Approximate Values And True Values

Throughout this chapter:

- y_n denotes a numerical approximation at x_n
- $y(x_n)$ denotes the true solution value
- $f_n = f(x_n, y_n)$ is a slope calculated from the numerical point
- $h = x_{n+1} - x_n$ is the constant step size

Do not silently replace y_n by the true value $y(x_n)$. A numerical method normally advances using its own approximate history.

Predictors And Correctors

A **predictor** gives a provisional endpoint value:

$$y_{n+1}^{(p)}.$$

Use that predicted height to calculate a predicted endpoint slope:

$$f_{n+1}^{(p)} = f\left(x_{n+1}, y_{n+1}^{(p)}\right).$$

A **corrector** then uses the new slope information to calculate an improved endpoint value y_{n+1}.

The superscript (p) means predicted. It is not a power.

Block 1 Summary

Numerical methods differ mainly in where they sample slopes and how they combine them.

Euler uses one old slope. Modified Euler predicts an endpoint and averages two slopes. RK4 creates four stages inside one interval. Multistep methods combine stored slopes from several intervals.

Modified Euler As A Predictor–Corrector Method

The Goal Of This Block

The goal is to derive modified Euler, distinguish the predicted and corrected values, and carry out two complete steps.

Predict With Euler

At (x_n, y_n), calculate:

$$f_n = f(x_n, y_n).$$

Euler predicts:

$$\boxed{y_{n+1}^{(p)} = y_n + hf_n}.$$

The next input is:

$$x_{n+1} = x_n + h.$$

Now calculate the slope at the predicted endpoint:

$$\boxed{f_{n+1}^{(p)} = f\left(x_{n+1}, y_{n+1}^{(p)}\right)}.$$

Correct With The Average Slope

Approximate the average slope across the interval by:

$$\frac{f_n + f_{n+1}^{(p)}}{2}.$$

Multiply this average by the step length h and add it to y_n:

$$\boxed{y_{n+1} = y_n + \frac{h}{2}\left(f_n + f_{n+1}^{(p)}\right)}.$$

Written as a complete algorithm:

$$\boxed{\begin{aligned} x_{n+1} &= x_n + h, \\ f_n &= f(x_n, y_n), \\ y_{n+1}^{(p)} &= y_n + hf_n, \\ f_{n+1}^{(p)} &= f\left(x_{n+1}, y_{n+1}^{(p)}\right), \\ y_{n+1} &= y_n + \frac{h}{2}\left(f_n + f_{n+1}^{(p)}\right). \end{aligned}}$$

This single-correction form is also called Heun's method.

Why The Average Is Better Than The Left Slope Alone

Euler uses a rectangular estimate for the accumulated change:

$$\Delta y \approx hf_n.$$

Modified Euler uses a trapezoidal estimate:

$$\Delta y \approx h\left(\frac{f_n + f_{n+1}^{(p)}}{2}\right).$$

The right-endpoint slope is not known in advance, so the predictor supplies a provisional point at which to estimate it.

A Two-Step Modified Euler Example

Approximate $y(0.4)$ for:

$$y' = 2x - y + 1, \qquad y(0) = 0,$$

using:

$$h = 0.2.$$

Here:

$$f(x, y) = 2x - y + 1, \qquad (x_0, y_0) = (0, 0).$$

Modified Euler Step 0 To Step 1

Calculate the left slope:

$$\begin{aligned} f_0 &= f(x_0, y_0) \\ &= f(0, 0) \\ &= 2(0) - 0 + 1 \\ &= 1. \end{aligned}$$

Update the input:

$$x_1 = x_0 + h = 0 + 0.2 = 0.2.$$

Predict the endpoint height:

$$\begin{aligned} y_1^{(p)} &= y_0 + hf_0 \\ &= 0 + 0.2(1) \\ &= 0.2. \end{aligned}$$

Calculate the predicted endpoint slope at $(x_1, y_1^{(p)}) = (0.2, 0.2)$:

$$\begin{aligned} f_1^{(p)} &= f\left(x_1, y_1^{(p)}\right) \\ &= f(0.2, 0.2) \\ &= 2(0.2) - 0.2 + 1 \\ &= 1.2. \end{aligned}$$

Correct using the average of f_0 and $f_1^{(p)}$:

$$\begin{aligned} y_1 &= y_0 + \frac{h}{2}\left(f_0 + f_1^{(p)}\right) \\ &= 0 + \frac{0.2}{2}(1 + 1.2) \\ &= 0.1(2.2) \\ &= 0.22. \end{aligned}$$

Thus:

$$\boxed{(x_1, y_1) = (0.2, 0.22)}.$$

The corrected value 0.22, not the predictor 0.2, begins the next step.

Modified Euler Step 1 To Step 2

Use the corrected point $(x_1, y_1) = (0.2, 0.22)$.

Calculate its slope:

$$\begin{aligned} f_1 &= f(x_1, y_1) \\ &= f(0.2, 0.22) \\ &= 2(0.2) - 0.22 + 1 \\ &= 1.18. \end{aligned}$$

Update the input:

$$x_2 = x_1 + h = 0.2 + 0.2 = 0.4.$$

Predict:

$$\begin{aligned} y_2^{(p)} &= y_1 + hf_1 \\ &= 0.22 + 0.2(1.18) \\ &= 0.456. \end{aligned}$$

Calculate the predicted endpoint slope at $(0.4, 0.456)$:

$$\begin{aligned} f_2^{(p)} &= f(0.4, 0.456) \\ &= 2(0.4) - 0.456 + 1 \\ &= 1.344. \end{aligned}$$

Correct:

$$\begin{aligned} y_2 &= y_1 + \frac{h}{2}\left(f_1 + f_2^{(p)}\right) \\ &= 0.22 + 0.1(1.18 + 1.344) \\ &= 0.22 + 0.1(2.524) \\ &= 0.4724. \end{aligned}$$

The second modified Euler step gives the endpoint estimate:

$$\boxed{y(0.4) \approx 0.4724}.$$

Compare With Euler And The Exact Value

With the same $h = 0.2$, two Euler steps give:

$$y_2^{\text{Euler}} = 0.44.$$

The exact solution is:

$$y(x) = 2x - 1 + e^{-x}.$$

At $x = 0.4$:

$$\begin{aligned} y(0.4) &= 2(0.4) - 1 + e^{-0.4} \\ &\approx 0.470320. \end{aligned}$$

The two absolute errors are:

$$|0.44 - 0.470320| \approx 0.030320,$$

and:

$$|0.4724 - 0.470320| \approx 0.002080.$$

For this step size and problem, the corrected average slope is substantially more accurate than Euler's left slope.

Block 2 Summary

Modified Euler performs:

$$\text{left slope} \to \text{Euler predictor} \to \text{predicted right slope} \to \text{average-slope correction}.$$

Never carry the predictor forward as though it were the corrected solution.

Classical Fourth-Order Runge–Kutta

The Goal Of This Block

The goal is to interpret the four RK4 stages and calculate one complete step without mixing slope and increment conventions.

The Increment Convention

This chapter defines each k_j as a proposed change in y, so every stage already includes the factor h:

$$\boxed{\begin{aligned} k_1 &= hf(x_n, y_n), \\ k_2 &= hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right), \\ k_3 &= hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right), \\ k_4 &= hf(x_n + h, y_n + k_3). \end{aligned}}$$

The update is:

$$\boxed{y_{n+1} = y_n + \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}}.$$

Also:

$$x_{n+1} = x_n + h.$$

Some books define k_j as slopes without h. Either convention is valid, but the two formulas must not be mixed.

Where The Four Stages Sample

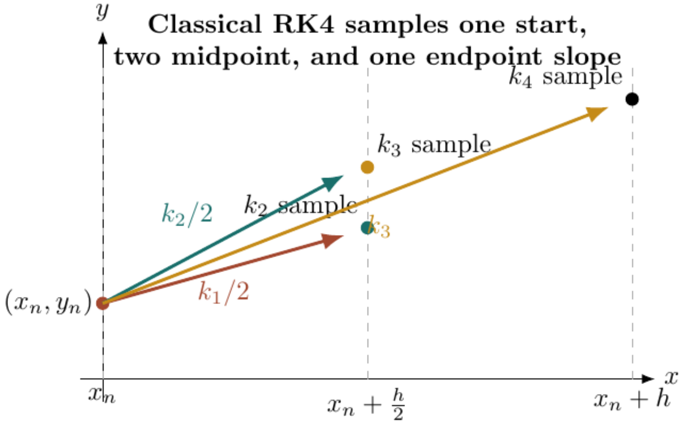

Figure 1: Schematic of the four classical RK4 stage locations

The stages represent:

- k_1: a start-point increment
- k_2: a midpoint increment predicted using k_1
- k_3: a second midpoint increment predicted using k_2
- k_4: an endpoint increment predicted using k_3

The two midpoint stages are not duplicates. Their estimated heights differ.

Why The Weights Are One, Two, Two, One

The final weighted increment is:

$$\frac{k_1 + 2k_2 + 2k_3 + k_4}{6}.$$

The weights sum to:

$$1 + 2 + 2 + 1 = 6.$$

Thus the formula behaves like a weighted average of four increments, with greater weight on the midpoint information.

The coefficients are chosen so that the Taylor expansion agrees with the true solution through terms of order h^4.

A Complete RK4 Step

Take one RK4 step for:

$$y' = y - x^2, \qquad y(0) = 0.5, \qquad h = 0.2.$$

The slope rule is:

$$f(x, y) = y - x^2.$$

The starting point is:

$$(x_0, y_0) = (0, 0.5).$$

Stage 1

Use the starting point:

$$\begin{aligned} k_1 &= hf(x_0, y_0) \\ &= 0.2f(0, 0.5) \\ &= 0.2\left(0.5 - 0^2\right) \\ &= 0.1. \end{aligned}$$

Stage 2

The midpoint input is:

$$x_0 + \frac{h}{2} = 0 + \frac{0.2}{2} = 0.1.$$

The first midpoint height estimate is:

$$y_0 + \frac{k_1}{2} = 0.5 + \frac{0.1}{2} = 0.55.$$

Substitute that midpoint:

$$\begin{aligned} k_2 &= 0.2f(0.1, 0.55) \\ &= 0.2\left(0.55 - (0.1)^2\right) \\ &= 0.2(0.54) \\ &= 0.108. \end{aligned}$$

Stage 3

The midpoint input remains 0.1. The second midpoint height uses k_2:

$$y_0 + \frac{k_2}{2} = 0.5 + \frac{0.108}{2} = 0.554.$$

Substitute:

$$\begin{aligned} k_3 &= 0.2f(0.1, 0.554) \\ &= 0.2\left(0.554 - (0.1)^2\right) \\ &= 0.2(0.544) \\ &= 0.1088. \end{aligned}$$

Stage 4

The endpoint input is:

$$x_0 + h = 0 + 0.2 = 0.2.$$

The endpoint height estimate uses k_3:

$$y_0 + k_3 = 0.5 + 0.1088 = 0.6088.$$

Substitute:

$$\begin{aligned} k_4 &= 0.2f(0.2, 0.6088) \\ &= 0.2\left(0.6088 - (0.2)^2\right) \\ &= 0.2(0.5688) \\ &= 0.11376. \end{aligned}$$

Combine The Four Stages

Use:

$$y_1 = y_0 + \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}.$$

Substitute all four increments:

$$\begin{aligned} y_1 &= 0.5 + \frac{0.1 + 2(0.108) + 2(0.1088) + 0.11376}{6} \\ &= 0.5 + \frac{0.64736}{6} \\ &\approx 0.607893. \end{aligned}$$

Combining the four weighted increments gives the RK4 endpoint estimate:

$$\boxed{y(0.2) \approx 0.607893}.$$

The exact solution is:

$$y(x) = x^2 + 2x + 2 - 1.5e^x.$$

At $x = 0.2$:

$$y(0.2) \approx 0.607896.$$

The one-step RK4 approximation agrees to about five decimal places.

Block 3 Summary

RK4 uses four linked stages. Each later stage must use the height estimate created by the specified earlier stage.

The update weights are:

$$1, \quad 2, \quad 2, \quad 1.$$

Keep the chosen k convention visible throughout the calculation.

Order, Error, And Computational Cost

The Goal Of This Block

The goal is to interpret method order as an error-scaling statement and compare improved accuracy with additional slope evaluations.

What Order Means

A method of global order p typically has error:

$$E(h) = O(h^p)$$

over a fixed interval, provided the solution is sufficiently smooth and the method is used in a stable regime.

The corresponding local truncation error is usually:

$$O(h^{p+1}).$$

The methods studied so far have global orders:

Method	Global order
Euler	1
Modified Euler	2
Classical RK4	4

What Step Halving Predicts

Assume the error is in the asymptotic regime and follows:

$$E(h) \approx Ch^p,$$

then:

$$E\left(\frac{h}{2}\right) \approx C\left(\frac{h}{2}\right)^p = \frac{E(h)}{2^p}.$$

Therefore halving h should reduce error by approximately:

- a factor of 2 for Euler
- a factor of 4 for modified Euler
- a factor of 16 for RK4

This comparison is asymptotic. It becomes reliable only when h is already small enough and roundoff or instability is not dominating.

An Original Error Comparison

For:

$$y' = 2x - y + 1, \qquad y(0) = 0,$$

the exact value at $x = 1$ is:

$$y(1) = 1 + e^{-1} \approx 1.367879441.$$

The absolute errors are:

h	Euler	Modified Euler	RK4
0.500	1.1788×10^{-1}	2.2746×10^{-2}	2.9140×10^{-4}
0.250	5.1473×10^{-2}	4.6496×10^{-3}	1.4758×10^{-5}
0.125	2.4271×10^{-2}	1.0538×10^{-3}	8.31×10^{-7}

The steeper RK4 line indicates faster error reduction as h decreases.

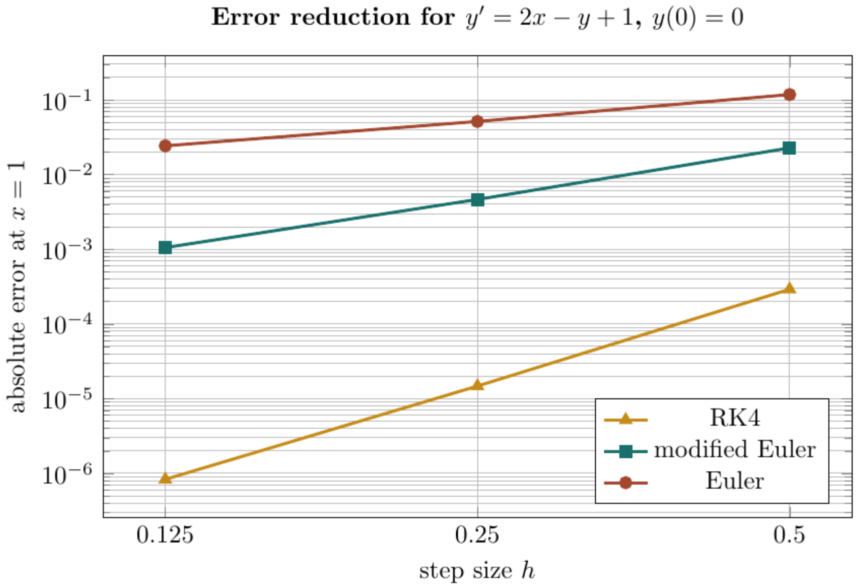

Figure 2: Log-scale comparison of Euler, modified Euler, and RK4 errors

Accuracy Is Not Free

Ignoring reusable work, one step needs approximately:

Method	New evaluations of f per step
Euler	1
Modified Euler	2
RK4	4

For a fixed h, RK4 costs more per step. However, its greater accuracy may permit a much larger h for the same error target.

The most efficient choice depends on the problem, desired tolerance, and cost of evaluating $f(x, y)$.

Estimating An Observed Order

Given errors E_h and $E_{h/2}$, estimate:

$$\boxed{p_{\text{obs}} = \log_2\left(\frac{E_h}{E_{h/2}}\right)}.$$

For example, if:

$$E_h = 0.008, \qquad E_{h/2} = 0.002,$$

then:

$$\begin{aligned} p_{\text{obs}} &= \log_2\left(\frac{0.008}{0.002}\right) \\ &= \log_2(4) \\ &= 2. \end{aligned}$$

The observed data are consistent with a second-order method.

Block 4 Summary

Order predicts how the global error scales when h is reduced. It does not guarantee accuracy for a large step, a nonsmooth solution, or an unstable calculation.

Compare methods using both error and work.

Why Multistep Methods Need A History

The Goal Of This Block

The goal is to understand what multistep formulas reuse, why the grid must be uniform, and where the first four solution values come from.

Store The Previous Slopes

Once y_j has been calculated, store:

$$f_j = f(x_j, y_j).$$

A four-step method can combine:

$$f_n, \qquad f_{n-1}, \qquad f_{n-2}, \qquad f_{n-3}.$$

This reuses old evaluations instead of constructing several new midpoint stages in every interval.

The indices describe time order. The largest index is the most recent value.

Why Equal Spacing Matters

The standard coefficients in ABM4 and Milne's method assume:

$$x_{j+1} - x_j = h$$

for every stored interval.

If the grid spacing changes, the fixed coefficients no longer represent the same interpolation or integration formula. Variable-step multistep methods need different coefficients.

The Start-Up Problem

A four-step formula cannot begin with only:

$$(x_0, y_0).$$

To calculate the first multistep prediction at x_4, it needs:

$$y_0, \quad y_1, \quad y_2, \quad y_3$$

and their slopes.

Use a one-step method such as RK4 to generate y_1, y_2, y_3. Then switch to the multistep formula.

The key intuition:

a method that uses history must first create that history.

A Shared Start-Up Table

The next two blocks use:

$$y' = 3x - y, \qquad y(0) = 1, \qquad h = 0.1.$$

Here:

$$f(x, y) = 3x - y.$$

RK4 supplies the following starting values:

j	x_j	y_j	$f_j = 3x_j - y_j$
0	0.0	1.000000000	−1.000000000
1	0.1	0.919350000	−0.619350000
2	0.2	0.874923606	−0.274923606
3	0.3	0.863273688	0.036726312

For example, the last slope is calculated locally as:

$$\begin{aligned} f_3 &= f(x_3, y_3) \\ &= 3(0.3) - 0.863273688 \\ &= 0.036726312. \end{aligned}$$

The first multistep target is:

$$x_4 = x_3 + h = 0.3 + 0.1 = 0.4.$$

Do Not Mix Start-Up Accuracy Casually

A fourth-order multistep method should be started with comparably accurate values. Poor Euler start-up values can contaminate later results even if the multistep formula itself is fourth order.

Using RK4 for start-up preserves the intended fourth-order accuracy under the usual smoothness and stability assumptions.

Block 5 Summary

ABM4 and Milne's method require four equally spaced solution values. Generate the missing start-up values with RK4, calculate and store their slopes, then begin the multistep method at $n = 3$ to obtain y_4.

Adams–Bashforth–Moulton Method

The Goal Of This Block

The goal is to apply the four-step Adams–Bashforth predictor and the Adams–Moulton corrector with every index and weighted term visible.

The AB4 Predictor

The explicit four-step Adams–Bashforth predictor is:

$$\boxed{y_{n+1}^{(p)} = y_n + \frac{h}{24}\left(55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}\right)}.$$

Read the history from newest to oldest:

$$f_n, \quad f_{n-1}, \quad f_{n-2}, \quad f_{n-3}.$$

For the first prediction, set $n = 3$:

$$\boxed{y_4^{(p)} = y_3 + \frac{h}{24}\left(55f_3 - 59f_2 + 37f_1 - 9f_0\right)}.$$

The AM4 Corrector

After predicting $y_{n+1}^{(p)}$, calculate:

$$f_{n+1}^{(p)} = f\left(x_{n+1}, y_{n+1}^{(p)}\right).$$

The Adams–Moulton corrector is:

$$\boxed{y_{n+1} = y_n + \frac{h}{24}\left(9f_{n+1}^{(p)} + 19f_n - 5f_{n-1} + f_{n-2}\right)}.$$

For the first ABM4 correction, set $n = 3$ and target y_4:

$$\boxed{y_4 = y_3 + \frac{h}{24}\left(9f_4^{(p)} + 19f_3 - 5f_2 + f_1\right)}.$$

Notice that the corrector does not use f_0. Its four slopes span the more recent range from f_1 through the predicted f_4.

ABM Step 1: Form The Predictor Bracket

Use the start-up slopes:

$$\begin{aligned}
f_0 &= -1.000000000,\\
f_1 &= -0.619350000,\\
f_2 &= -0.274923606,\\
f_3 &= 0.036726312.
\end{aligned}$$

Substitute each weighted term:

$$\begin{aligned}
55f_3 &= 55(0.036726312) = 2.019947160,\\
-59f_2 &= -59(-0.274923606) = 16.220492732,\\
37f_1 &= 37(-0.619350000) = -22.915950000,\\
-9f_0 &= -9(-1.000000000) = 9.000000000.
\end{aligned}$$

Add the four terms:

$$\begin{aligned}
B_{\mathrm{AB}} &= 55f_3 - 59f_2 + 37f_1 - 9f_0\\
&= 2.019947160 + 16.220492732 - 22.915950000 + 9.000000000\\
&= 4.324489892.
\end{aligned}$$

ABM Step 2: Calculate The Predicted Value And Slope

Use:

$$y_4^{(p)} = y_3 + \frac{h}{24} B_{\text{AB}}.$$

Substitute $y_3 = 0.863273688$, $h = 0.1$, and $B_{\text{AB}} = 4.324489892$:

$$\begin{aligned} y_4^{(p)} &= 0.863273688 + \frac{0.1}{24}(4.324489892) \\ &\approx 0.881292397. \end{aligned}$$

The endpoint input is:

$$x_4 = 0.4.$$

Calculate the predicted endpoint slope from $f(x, y) = 3x - y$:

$$\begin{aligned} f_4^{(p)} &= f\left(x_4, y_4^{(p)}\right) \\ &= 3(0.4) - 0.881292397 \\ &\approx 0.318707603. \end{aligned}$$

The corrector requires this new slope, not the old f_3.

ABM Step 3: Apply The Corrector

Form the corrector bracket:

$$B_{\mathrm{AM}} = 9f_4^{(p)} + 19f_3 - 5f_2 + f_1.$$

For the AM4 bracket $B_{\mathrm{AM}} = 9f_4^{(p)} + 19f_3 - 5f_2 + f_1$, calculate each weighted term:

$$\begin{aligned}
9f_4^{(p)} &\approx 9(0.318707603) = 2.868368427,\\
19f_3 &= 19(0.036726312) = 0.697799928,\\
-5f_2 &= -5(-0.274923606) = 1.374618030,\\
f_1 &= -0.619350000.
\end{aligned}$$

Add:

$$\begin{aligned}
B_{\mathrm{AM}} &\approx 2.868368427 + 0.697799928 + 1.374618030 - 0.619350000\\
&\approx 4.321436385.
\end{aligned}$$

Now correct:

$$\begin{aligned}
y_4 &= y_3 + \frac{h}{24}B_{\mathrm{AM}}\\
&= 0.863273688 + \frac{0.1}{24}(4.321436385)\\
&\approx 0.881279673.
\end{aligned}$$

The corrected ABM4 estimate at the target input $x = 0.4$ is:

$$\boxed{y(0.4) \approx 0.881279673}.$$

Compare Predictor, Corrector, And Exact Value

The exact solution is:

$$y(x) = 3x - 3 + 4e^{-x}.$$

At $x = 0.4$:

$$y(0.4) \approx 0.881280184.$$

The values are:

$$\begin{aligned} y_4^{(p)} &\approx 0.881292397, \\ y_4 &\approx 0.881279673, \\ y(0.4) &\approx 0.881280184. \end{aligned}$$

The correction moves the prediction much closer to the exact value.

Block 6 Summary

ABM4 uses old slopes to predict, evaluates a new predicted slope at the endpoint, then corrects with a different weighted combination.

The essential chain is:

$$\{f_0, f_1, f_2, f_3\} \to y_4^{(p)} \to f_4^{(p)} \to y_4.$$

Milne's Method, Diagnostics, And Common Mistakes

The Goal Of This Block

The goal is to apply Milne's predictor and corrector to the same start-up data, then use predictor–corrector agreement as a diagnostic rather than a proof.

Milne's Predictor

Milne's four-step predictor is:

$$\boxed{y_{n+1}^{(p)} = y_{n-3} + \frac{4h}{3}\left(2f_n - f_{n-1} + 2f_{n-2}\right)}.$$

For the first Milne prediction, set $n = 3$ and target $y_4^{(p)}$:

$$\boxed{y_4^{(p)} = y_0 + \frac{4h}{3}\left(2f_3 - f_2 + 2f_1\right)}.$$

Unlike AB4, the base value is y_{n-3}, not y_n.

Milne's Corrector

First calculate:

$$f_{n+1}^{(p)} = f\left(x_{n+1}, y_{n+1}^{(p)}\right).$$

Then use the Simpson-type corrector:

$$\boxed{y_{n+1} = y_{n-1} + \frac{h}{3}\left(f_{n-1} + 4f_n + f_{n+1}^{(p)}\right)}.$$

For the first Milne correction, set $n = 3$ and target y_4:

$$\boxed{y_4 = y_2 + \frac{h}{3}\left(f_2 + 4f_3 + f_4^{(p)}\right)}.$$

The base value is now y_2.

A Complete Milne Prediction

Use the same problem:

$$y' = 3x - y, \qquad h = 0.1.$$

Form the predictor bracket:

$$\begin{aligned} B_{\mathrm{M}} &= 2f_3 - f_2 + 2f_1 \\ &= 2(0.036726312) - (-0.274923606) + 2(-0.619350000) \\ &= 0.073452624 + 0.274923606 - 1.238700000 \\ &= -0.890323770. \end{aligned}$$

Use $y_0 = 1$:

$$\begin{aligned} y_4^{(p)} &= y_0 + \frac{4h}{3} B_{\mathrm{M}} \\ &= 1 + \frac{4(0.1)}{3}(-0.890323770) \\ &\approx 0.881290164. \end{aligned}$$

Calculate The Predicted Slope And Correct

At $x_4 = 0.4$, calculate:

$$\begin{aligned} f_4^{(p)} &= 3x_4 - y_4^{(p)} \\ &= 3(0.4) - 0.881290164 \\ &\approx 0.318709836. \end{aligned}$$

Form the corrector bracket:

$$\begin{aligned} B_{\text{MC}} &= f_2 + 4f_3 + f_4^{(p)} \\ &= -0.274923606 + 4(0.036726312) + 0.318709836 \\ &= -0.274923606 + 0.146905248 + 0.318709836 \\ &\approx 0.190691478. \end{aligned}$$

Use the base value $y_2 = 0.874923606$:

$$\begin{aligned} y_4 &= y_2 + \frac{h}{3}B_{\text{MC}} \\ &= 0.874923606 + \frac{0.1}{3}(0.190691478) \\ &\approx 0.881279988. \end{aligned}$$

The completed Milne correction gives:

$$\boxed{y(0.4) \approx 0.881279988}.$$

The exact value is approximately 0.881280184.

The Predictor–Corrector Gap

For Milne's calculation:

$$\begin{aligned}\left|y_4 - y_4^{(p)}\right| &\approx |0.881279988 - 0.881290164| \\ &\approx 1.02 \times 10^{-5}.\end{aligned}$$

A small gap is reassuring. A large gap can indicate:

- a step size that is too large
- numerical instability
- inaccurate start-up values
- an indexing or arithmetic error
- rapid variation or a nearby singularity

The gap is a diagnostic, not a guaranteed bound on the true error.

Common Mistake: Skipping The Predicted Derivative

The corrector needs:

$$f_{n+1}^{(p)} = f\left(x_{n+1}, y_{n+1}^{(p)}\right).$$

Do not insert $y_{n+1}^{(p)}$ directly where a slope belongs, and do not reuse f_n as though it were the endpoint slope.

Common Mistake: Shifting The History

For AB4 at $n = 3$, the slope order is:

$$f_3, \quad f_2, \quad f_1, \quad f_0.$$

The coefficient 55 multiplies the newest slope f_3; the coefficient -9 multiplies the oldest slope f_0.

Write a small indexed table before substitution.

Common Mistake: Using The Wrong Base Value

The formulas start from different values:

- AB4 predictor and AM4 corrector start from y_n
- Milne predictor starts from y_{n-3}
- Milne corrector starts from y_{n-1}

At $n = 3$, these are y_3, y_0, and y_2, respectively.

Common Mistake: Using Unequal Grid Spacing

The displayed ABM4 and Milne coefficients assume one constant h. Before using stored values, verify:

$$x_1 - x_0 = x_2 - x_1 = x_3 - x_2 = h.$$

If the spacings differ, use a variable-step formula or restart with an appropriate one-step method.

Common Mistake: Rounding The Stored History Too Early

Each rounded y_j changes f_j, and each changed slope is multiplied by a large positive or negative coefficient.

Retain guard digits in start-up values, slopes, predictions, and corrections. Round the final reported value after the step is complete.

A Reliable Method Checklist

For Modified Euler.

1. calculate f_n
2. predict $y_{n+1}^{(p)}$
3. calculate $f_{n+1}^{(p)}$
4. average the two slopes
5. carry the corrected y_{n+1} forward

For RK4.

1. state whether each k_j is a slope or an increment
2. show each stage point
3. use k_1 for stage 2, k_2 for stage 3, and k_3 for stage 4
4. apply the weights $1, 2, 2, 1$

For A Four-Step Predictor–Corrector.

1. verify equal spacing
2. create y_1, y_2, y_3 with RK4
3. tabulate f_0, f_1, f_2, f_3
4. predict $y_4^{(p)}$
5. calculate $f_4^{(p)}$
6. correct to obtain y_4
7. compare the predictor and corrector

Block 7 Summary

Milne and ABM4 are fourth-order predictor–corrector methods with different history weights and base values.

Their formulas are compact, but a safe calculation expands every indexed term before adding it.

Original Practice Set

How To Use These Problems

For every method, write the formula before inserting numbers. Keep predicted values marked with (p) and keep RK4 stage inputs visible.

For multistep problems, copy the history into an indexed table before applying weights.

Problems 1 To 4: Modified Euler And RK4

Problem 1: One Modified Euler Step. For:

$$y' = x + y^2, \qquad y(0) = 0.5,$$

use modified Euler with $h = 0.1$ to find y_1.

Problem 2: Two Modified Euler Steps. For:

$$y' = 1 - y, \qquad y(0) = 0,$$

use modified Euler with $h = 0.2$ to approximate $y(0.4)$.

Problem 3: Diagnose A Predictor Error. A learner predicts $y_{n+1}^{(p)}$ correctly but calculates the endpoint slope as:

$$f\left(x_n, y_{n+1}^{(p)}\right).$$

Explain the error and state the correct endpoint slope.

Problem 4: One Complete RK4 Step. For:

$$y' = 3x - 2y, \qquad y(0) = 1,$$

take one classical RK4 step with $h = 0.2$ using the increment convention.

Problems 5 To 8: Stages, Work, Order, And ABM4

Problem 5: Identify Four RK4 Stage Points. At $(x_n, y_n) = (1, 2)$ with $h = 0.2$, suppose:

$$k_1 = 0.6, \qquad k_2 = 0.65, \qquad k_3 = 0.67.$$

State the (x, y) point at which each of k_1, k_2, k_3, k_4 evaluates f.

Problem 6: Count Function Evaluations. A calculation uses 20 equal steps. Ignoring reusable endpoint slopes, how many evaluations of f are required by Euler, modified Euler, and RK4?

Problem 7: Estimate An Observed Order. A method has target errors:

$$E_h = 0.008, \qquad E_{h/2} = 0.002.$$

Estimate its observed order.

Problem 8: Carry Out One ABM4 Step. For $y' = x + y$ with $h = 0.1$, RK4 has supplied:

j	x_j	y_j	$f_j = x_j + y_j$
0	0.0	1.000000000	1.000000000
1	0.1	1.110341667	1.210341667
2	0.2	1.242805142	1.442805142
3	0.3	1.399716994	1.699716994

Use AB4 and AM4 once to approximate $y(0.4)$.

Problems 9 To 12: Milne And Diagnostics

Problem 9: Carry Out One Milne Step. Use the same $y' = x + y$ history from Problem 8 to calculate one Milne prediction and correction at $x_4 = 0.4$.

Problem 10: Explain The Start-Up Requirement. Why can neither ABM4 nor Milne's method begin directly from only $y(x_0) = y_0$? State a suitable start-up procedure.

Problem 11: Respond To A Large Correction. A predictor gives $y_{n+1}^{(p)} = 1.25$, while the corrector gives $y_{n+1} = 1.80$. What should be checked before accepting the step?

Problem 12: Detect An Invalid Grid. Can the standard ABM4 formula be applied directly to values at:

$$x_0 = 0, \quad x_1 = 0.1, \quad x_2 = 0.25, \quad x_3 = 0.35?$$

Explain the decision.

Worked Answers: Problems 1 To 4

Answer 1. The data are:

$$f(x,y) = x + y^2, \qquad (x_0, y_0) = (0, 0.5), \qquad h = 0.1.$$

Calculate the left slope:

$$\begin{aligned} f_0 &= f(0, 0.5) \\ &= 0 + (0.5)^2 \\ &= 0.25. \end{aligned}$$

Update the input:

$$x_1 = x_0 + h = 0.1.$$

Predict:

$$\begin{aligned} y_1^{(p)} &= y_0 + hf_0 \\ &= 0.5 + 0.1(0.25) \\ &= 0.525. \end{aligned}$$

Calculate the predicted endpoint slope:

$$\begin{aligned} f_1^{(p)} &= f(0.1, 0.525) \\ &= 0.1 + (0.525)^2 \\ &= 0.1 + 0.275625 \\ &= 0.375625. \end{aligned}$$

Use the left and predicted endpoint slopes to correct the first step:

$$\begin{aligned} y_1 &= y_0 + \frac{h}{2}\left(f_0 + f_1^{(p)}\right) \\ &= 0.5 + 0.05(0.25 + 0.375625) \\ &= 0.5 + 0.03128125 \\ &= 0.53128125. \end{aligned}$$

Therefore:

$$\boxed{y(0.1) \approx 0.53128125}.$$

Answer 2. Use:

$$f(x,y) = 1 - y, \qquad (x_0, y_0) = (0,0), \qquad h = 0.2.$$

For the first step, calculate:

$$f_0 = 1 - y_0 = 1.$$

Predict:

$$y_1^{(p)} = 0 + 0.2(1) = 0.2.$$

At $x_1 = 0.2$, calculate:

$$f_1^{(p)} = 1 - 0.2 = 0.8.$$

Use the first-step slopes to calculate the corrected value:

$$\begin{aligned} y_1 &= 0 + \frac{0.2}{2}(1 + 0.8) \\ &= 0.18. \end{aligned}$$

Second Modified Euler Step. Begin the second step with the corrected $y_1 = 0.18$:

$$f_1 = 1 - 0.18 = 0.82.$$

Predict:

$$\begin{aligned} y_2^{(p)} &= y_1 + hf_1 \\ &= 0.18 + 0.2(0.82) \\ &= 0.344. \end{aligned}$$

At $x_2 = 0.4$, the predicted endpoint slope is:

$$f_2^{(p)} = 1 - 0.344 = 0.656.$$

Use the second-step slopes to calculate the corrected value:

$$\begin{aligned} y_2 &= 0.18 + \frac{0.2}{2}(0.82 + 0.656) \\ &= 0.18 + 0.1(1.476) \\ &= 0.3276. \end{aligned}$$

After two corrected steps, the modified Euler estimate is:

$$\boxed{y(0.4) \approx 0.3276}.$$

Answer 3. The predicted height belongs to the new input:

$$x_{n+1} = x_n + h.$$

Therefore the predicted endpoint slope must be:

$$\boxed{f_{n+1}^{(p)} = f\left(x_{n+1}, y_{n+1}^{(p)}\right)}.$$

Using x_n with $y_{n+1}^{(p)}$ mixes the old input and predicted new height.

Answer 4. Use:

$$f(x,y) = 3x - 2y, \qquad (x_0, y_0) = (0,1), \qquad h = 0.2.$$

Stage 1:

$$\begin{aligned} k_1 &= 0.2f(0,1) \\ &= 0.2(3(0) - 2(1)) \\ &= -0.4. \end{aligned}$$

Stage 2 uses $(0.1, 1 + k_1/2) = (0.1, 0.8)$:

$$\begin{aligned} k_2 &= 0.2f(0.1, 0.8) \\ &= 0.2(3(0.1) - 2(0.8)) \\ &= 0.2(-1.3) \\ &= -0.26. \end{aligned}$$

Stage 3 uses $(0.1, 1 + k_2/2) = (0.1, 0.87)$:

$$\begin{aligned} k_3 &= 0.2f(0.1, 0.87) \\ &= 0.2(3(0.1) - 2(0.87)) \\ &= 0.2(-1.44) \\ &= -0.288. \end{aligned}$$

Stage 4 uses $(0.2, 1 + k_3) = (0.2, 0.712)$:

$$\begin{aligned} k_4 &= 0.2f(0.2, 0.712) \\ &= 0.2(3(0.2) - 2(0.712)) \\ &= 0.2(-0.824) \\ &= -0.1648. \end{aligned}$$

Combine:

$$\begin{aligned} y_1 &= 1 + \frac{-0.4 + 2(-0.26) + 2(-0.288) - 0.1648}{6} \\ &= 1 - \frac{1.6608}{6} \\ &= 0.7232. \end{aligned}$$

Therefore the one-step RK4 estimate is:

$$\boxed{y(0.2) \approx 0.7232}.$$

Worked Answers: Problems 5 To 8

Answer 5. With the increment convention, RK4 samples:

$$k_1 = hf(x_n, y_n),$$

so the k_1 point is:

$$\boxed{(1,2)}.$$

For k_2, use:

$$\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right).$$

Substitute $h = 0.2$ and $k_1 = 0.6$:

$$\boxed{(1.1, 2.3)}.$$

For k_3, use k_2 in the midpoint height:

$$\left(1.1, 2 + \frac{0.65}{2}\right) = \boxed{(1.1, 2.325)}.$$

For k_4, use the full endpoint and k_3:

$$(1.2, 2 + 0.67) = \boxed{(1.2, 2.67)}.$$

Answer 6. For 20 steps:

- Euler uses 1 evaluation per step
- modified Euler uses 2 evaluations per step
- RK4 uses 4 evaluations per step

The corrected Milne estimate at the target input $x = 0.4$ is:

$$\begin{aligned} N_{\text{Euler}} &= 20(1) = 20, \\ N_{\text{ME}} &= 20(2) = 40, \\ N_{\text{RK4}} &= 20(4) = 80. \end{aligned}$$

Answer 7. Use:

$$p_{\text{obs}} = \log_2\left(\frac{E_h}{E_{h/2}}\right).$$

Substitute:

$$\begin{aligned} p_{\text{obs}} &= \log_2\left(\frac{0.008}{0.002}\right) \\ &= \log_2(4) \\ &= 2. \end{aligned}$$

The observed errors are consistent with a second-order method.

Answer 8. For the first ABM4 step, $n = 3$ and $x_4 = 0.4$.

Use the predictor:

$$y_4^{(p)} = y_3 + \frac{0.1}{24}\left(55f_3 - 59f_2 + 37f_1 - 9f_0\right).$$

Substitute the table values:

$$\begin{aligned} y_4^{(p)} &= 1.399716994 + \frac{0.1}{24}[55(1.699716994) - 59(1.442805142) \\ &\quad + 37(1.210341667) - 9(1.000000000)] \\ &\approx 1.583640215. \end{aligned}$$

For $f(x, y) = x + y$, the predicted endpoint slope is:

$$\begin{aligned} f_4^{(p)} &= f(0.4, 1.583640215) \\ &= 0.4 + 1.583640215 \\ &= 1.983640215. \end{aligned}$$

Apply the corrector:

$$y_4 = y_3 + \frac{0.1}{24}\left(9f_4^{(p)} + 19f_3 - 5f_2 + f_1\right).$$

Substitute:

$$\begin{aligned} y_4 &= 1.399716994 + \frac{0.1}{24}[9(1.983640215) + 19(1.699716994) \\ &\quad - 5(1.442805142) + 1.210341667] \\ &\approx 1.583649081. \end{aligned}$$

Thus:

$$\boxed{y(0.4) \approx 1.583649081}.$$

Worked Answers: Problems 9 To 12

Answer 9. Use Milne's predictor at $n = 3$:

$$y_4^{(p)} = y_0 + \frac{4(0.1)}{3}\left(2f_3 - f_2 + 2f_1\right).$$

Substitute the history:

$$\begin{aligned} y_4^{(p)} &= 1 + \frac{0.4}{3}[2(1.699716994) - 1.442805142 \\ &\quad + 2(1.210341667)] \\ &\approx 1.583641624. \end{aligned}$$

Calculate the predicted endpoint slope:

$$\begin{aligned} f_4^{(p)} &= f(0.4, 1.583641624) \\ &= 0.4 + 1.583641624 \\ &= 1.983641624. \end{aligned}$$

Use Milne's corrector:

$$y_4 = y_2 + \frac{0.1}{3}\left(f_2 + 4f_3 + f_4^{(p)}\right).$$

Substitute:

$$\begin{aligned} y_4 &= 1.242805142 + \frac{0.1}{3}[1.442805142 + 4(1.699716994) \\ &\quad + 1.983641624] \\ &\approx 1.583648966. \end{aligned}$$

The corrected Milne estimate at the target input $x = 0.4$ is:

$$\boxed{y(0.4) \approx 1.583648966}.$$

Answer 10. ABM4 and Milne's method use several earlier values and slopes. At the first multistep calculation they require:

$$y_0, \quad y_1, \quad y_2, \quad y_3$$

and:

$$f_0, \quad f_1, \quad f_2, \quad f_3.$$

The initial condition supplies only y_0. A suitable procedure is:

1. use RK4 with the intended step size to calculate y_1, y_2, y_3
2. evaluate $f_j = f(x_j, y_j)$ at each start-up point
3. begin the four-step method with $n = 3$

Answer 11. The predictor–corrector gap is:

$$|1.80 - 1.25| = 0.55,$$

which is large relative to the values themselves.

Before accepting the step:

1. check coefficient signs and history indices
2. verify the predicted endpoint derivative
3. inspect start-up accuracy and rounding
4. repeat with a smaller step size
5. consider rapid variation, stiffness, or a nearby singularity

A corrector is intended to refine the prediction. A large change requires investigation.

Answer 12. Calculate the spacings:

$$\begin{aligned} x_1 - x_0 &= 0.1, \\ x_2 - x_1 &= 0.15, \\ x_3 - x_2 &= 0.10. \end{aligned}$$

They are not equal. Therefore no single constant h describes this history.

The standard fixed-step ABM4 coefficients cannot be applied directly:

$$\boxed{\text{the grid is invalid for the displayed fixed-step formula.}}$$

Use a variable-step method, construct an equally spaced history, or restart with a suitable one-step method.

Chapter 19 Final Summary

Modified Euler predicts with Euler and corrects with the average of the left and predicted right slopes:

$$\boxed{\begin{aligned} y_{n+1}^{(p)} &= y_n + hf_n, \\ f_{n+1}^{(p)} &= f\left(x_{n+1}, y_{n+1}^{(p)}\right), \\ y_{n+1} &= y_n + \frac{h}{2}\left(f_n + f_{n+1}^{(p)}\right). \end{aligned}}$$

Classical RK4 uses four increments:

$$\boxed{\begin{aligned} k_1 &= hf(x_n, y_n), \\ k_2 &= hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right), \\ k_3 &= hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right), \\ k_4 &= hf(x_n + h, y_n + k_3), \end{aligned}}$$

RK4 combines these four increments using:

$$\boxed{y_{n+1} = y_n + \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}}.$$

Euler, modified Euler, and RK4 have global orders 1, 2, and 4, respectively.

Four-step multistep methods require RK4 start-up values. ABM4 uses:

$$\boxed{y_{n+1}^{(p)} = y_n + \frac{h}{24}\left(55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}\right)}$$

followed by:

$$\boxed{y_{n+1} = y_n + \frac{h}{24}\left(9f_{n+1}^{(p)} + 19f_n - 5f_{n-1} + f_{n-2}\right)}.$$

Milne's method uses:

$$\boxed{y_{n+1}^{(p)} = y_{n-3} + \frac{4h}{3}\left(2f_n - f_{n-1} + 2f_{n-2}\right)}$$

and:

$$\boxed{y_{n+1} = y_{n-1} + \frac{h}{3}\left(f_{n-1} + 4f_n + f_{n+1}^{(p)}\right)}.$$

The chapter's central lesson is:

identify exactly where every slope is sampled, preserve the time indices, generate trustworthy start-up values, and treat predictor–corrector disagreement as a warning that deserves investigation.